

GADDE BHANU PRAKASH

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PROFESSIONAL SUMMARY

AI Engineer & Data Scientist with hands-on experience architecting end-to-end machine learning systems—from data ingestion and experimentation to deployment and monitoring at scale. Specialized in **Deep Learning, Time-Series Forecasting, and Generative AI**. Proven track record of shipping production-grade AI solutions that deliver measurable business impact. Passionate about building autonomous, self-healing AI systems and bridging the gap between research and production.

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript
ML/DL	PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, PyOD, Autoencoders, Isolation Forest
GenAI & Agents	LangChain, LangGraph, Google Gemini API, Ollama (Mistral), Groq API (Llama 3.1)
Data Engineering	Great Expectations, SQLAlchemy, Pandas, NumPy, PostgreSQL (Neon), Prompt Engineering
Backend & Frontend	FastAPI, Streamlit, Flask, Docker
DevOps & Cloud	Docker, Kubernetes, Render, Azure, GitHub Actions, CI/CD
Other	SHAP, MLflow, Git, cron-job.org

PROJECTS

Project VIGIL — Autonomous AI Data Quality & Observability Platform PyTorch · FastAPI · Streamlit · Great Expectations · LangChain · Gemini · Docker · Render	Live Demo  GitHub
- Built an autonomous AI system that continuously monitors data quality, detects hidden anomalies using Isolation Forest + PyTorch Autoencoder, and auto-generates corrective SQL using LangChain + Google Gemini. - Implemented 5 automated validation rules using Great Expectations with statistical drift detection (KS-Test, Chi-Squared) for early pipeline failure alerts. - Designed a safety critic agent that reviews and blocks dangerous SQL commands (DELETE, DROP, TRUNCATE), ensuring production-safe autonomous healing. - Achieved a unified Data Trust Score (0-100%) combining GX validation (20%) + ML anomaly detection (4.36% anomalies) into a single executive-friendly metric. - Deployed full-stack microservices with FastAPI, Streamlit, Docker, and CI/CD on Render and Streamlit Cloud with 24/7 uptime via cron-job.org pings.	
SynergyOS — Cognitive Supply Chain Twin Prophet · LightGBM · Google OR-Tools · LangChain · Ollama · FastAPI · Streamlit	Live Demo  GitHub
- Engineered a hybrid time-series forecasting pipeline (Prophet + LightGBM) achieving hyper-local demand sensing across 5+ warehouses with a 4-week prediction horizon, enabling proactive inventory planning. - Designed a multi-objective inventory optimization engine using Google OR-Tools (Linear Programming) minimizing holding, stockout, and transfer costs simultaneously. - Built a GenAI Co-Pilot with LangChain + Ollama (Mistral 7B) converting complex optimization outputs into 1-line actionable alerts, reducing manager decision time by 80%. - Integrated real-time Weather API data as external features to enhance demand forecasting accuracy, deployed full-stack via FastAPI + Streamlit.	
Bank Customer Churn Predictor XGBoost · SHAP · Scikit-learn · Streamlit · Docker	Live Demo  GitHub
- Developed an XGBoost classification model on 50,000+ customer records, handling class imbalance with SMOTE and achieving 0.89 AUC-ROC for early churn risk identification. - Integrated SHAP (SHapley Additive exPlanations) to provide transparent, feature-level explanations for each prediction, building trust with business stakeholders. - Deployed a live interactive Streamlit web app with real-time prediction and SHAP force plots, containerized with Docker and hosted on Streamlit Cloud.	
Deep Learning-Driven Visual Search & Image Recommendation CNNs · Autoencoders · Siamese Networks · GANs · Attention Networks	 GitHub
- Conducted a comprehensive survey and implementation of Content-Based Image Retrieval (CBIR) using deep learning architectures spanning 2011–2020. - Systematically categorized approaches across supervision types, network typologies, and descriptor methods. - Implemented binary descriptors (deep hashing) for sub-linear lookups in massive image datasets.	

EDUCATION

PBR Visvodaya Institute of Technology and Science <i>B.Tech in Computer Science (Artificial Intelligence)</i>	2021 – 2025 CGPA: 8.0/10.0
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ACHIEVEMENTS & CERTIFICATIONS

- Participated in **Meghaton** at IIIT Hyderabad — Web Design Competition at PBR VITS
- Completed multiple end-to-end AI/ML projects with production-grade deployments